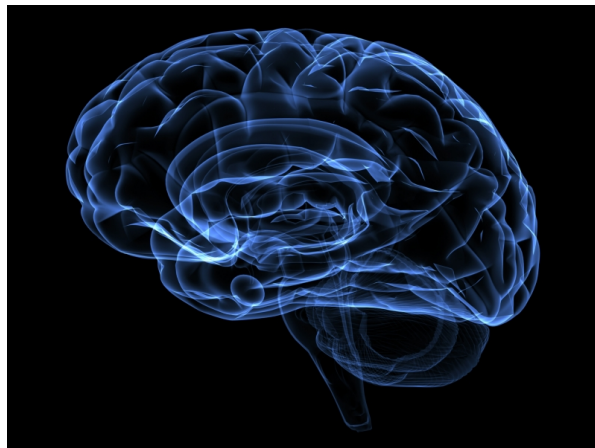




Pediatric Neuromusculoskeletal Disorders Part 1



Tami Hendriksz, DO, FACOP, FAAP
TOURO UNIVERSITY CALIFORNIA
COLLEGE OF OSTEOPATHIC MEDICINE

Pediatric Musculoskeletal Disorders Part 1

Table of Contents

Section Title	Page
<u>Objectives</u>	3
<u>CNS Development</u>	3
<u>Congenital Anomalies of the Nervous System</u>	4
<u>Neurologic Evaluation</u>	5
<u>Physiology of Intracranial Pressure</u>	6
<u>Increased Intracranial Pressure</u>	7
<u>Psychomotor Retardation</u>	10
<u>Disorders Involving Weakness</u>	12
<u>Neurocutaneous Disorders</u>	16
<u>References</u>	23

The Section Titles above serve as hyperlinks to the corresponding portions of the document for quick reference.

Objectives

At the end of this module the learner should be able to:

- Recognize the range of anomalies associated with spinal dysraphism (spina bifida occulta, meningocele & myelomeningocele) and their clinical manifestations
- Identify the major symptoms and common causes of increased intracranial pressure (ICP) (specifically, Pseudotumor Cerebri, Meningitis and Hydrocephalus)
- Recognize the specific clinical features of Spinal Muscular Atrophy, Duchenne's Musculodystrophy, Myotonic Dystrophy and Guillan-Barre Syndrome
- Distinguish between the presented neurocutaneous disorders (TS, NF1, NF2, SW) and recognize their clinical features

CNS Development

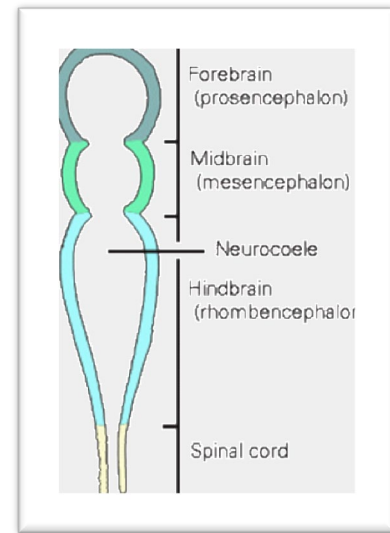
How does the CNS Develop?

The neural plate of the embryonic ectoderm is the precursor of the nervous system.

The neural plate develops at **18 days** of gestation and gives rise to the **neural tube and neural crest cells**. The **neural tube** forms the brain and spinal cord, while the **neural crest cells** form the peripheral nervous system, meninges, melanocytes, and adrenal medulla.

Neural Tube:

The neural tube begins to form at 22 days of gestation. The **rostral end** forms the **brain**, the **caudal** region forms the **spinal cord**, and the lumen of the neural tube forms the brain's ventricles and central canal of the spinal cord.



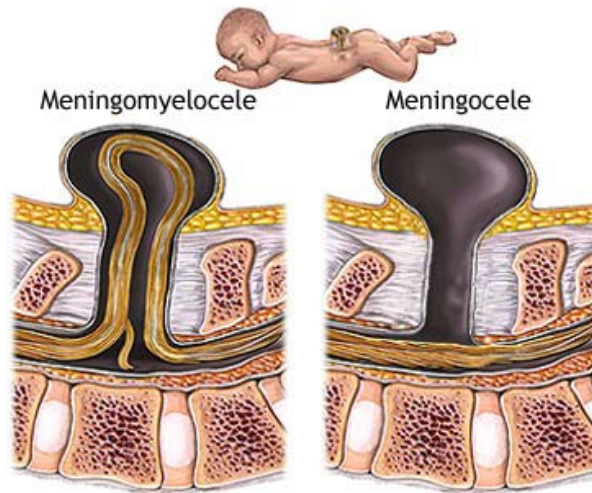
Congenital Anomalies of the Nervous System

Congenital anomalies can arise because of **spinal dysraphism** and **folate deficiencies**.

Spinal dysraphism or neural tube defects are defective closures of the caudal neural tube. There is a range of anomalies:

- (1) Defects of L5 or S1 vertebral arches
- (2) **Spina bifida occulta** – defects of underlying bone or canal (overlying skin is intact)
May present as a sacral dimple, or thick tuft of hair overlying the lumbosacral area
- (3) **Meningocele** – sac filled with CSF due to protrusion of the dura mater
- (4) **Myelomeningocele** – spinal canal and meninges exposed





Folate is essential for the normal development of the fetal nervous system because of its ability to prevent elevated levels of homocysteine in amniotic fluid. A deficiency of folate would prevent the normal process from occurring.

Neurologic Evaluation

The neurologic evaluation of pediatric patient varies with age.

In **infants** it is important to examine:

- Hair, skin, teeth, and nails
- Head circumference
- Fontanel
- Primitive reflexes (Moro, rooting, such, etc)
- Posture
- Movement and tone

Note: *abnormalities of the ectoderm (hair, skin, etc) may indicate abnormalities in the CNS due to the embryonic origins*

At **all ages**, we examine:

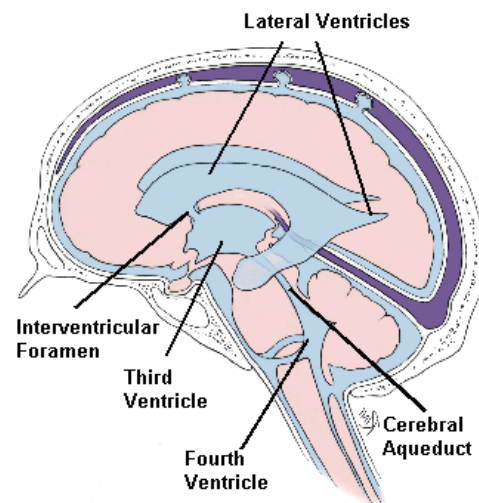
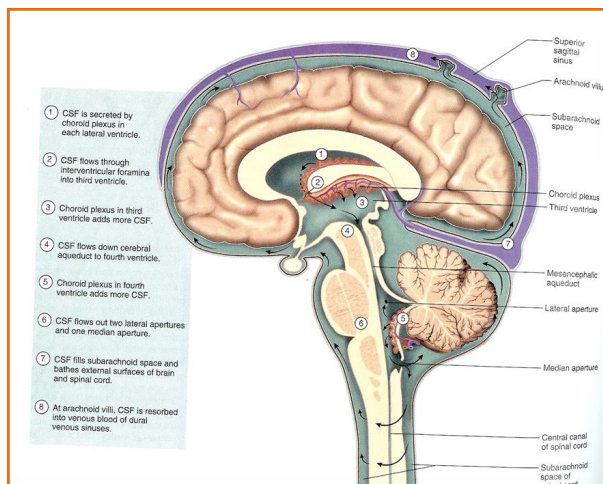
- Cranial Nerves
- Motor Exam: strength, tone, coordination, gait, reflexes, involuntary movements
- Sensory exam
- Mental status
- Hair, skin, teeth, and nails

Physiology of Intracranial Pressure

The skull is a **rigid container** which contains **brain (80-85%), blood (5-10%), and CSF (10-15%)**. An imbalance of any of these will lead to increased intracranial pressure.

Flow of CSF

The CSF flows from the lateral ventricle → foramen of Monroe → 3rd ventricle → Cerebral aqueduct → 4th ventricle → foramen of Magendie and Luschka



A blockage anywhere along this pathway can lead to increased CSF and hydrocephalus (and corresponding increased intracranial pressure).

Increased Intracranial Pressure

Pathophysiology of increased ICP

Anything that takes up more space in the rigid container that is the skull can lead to increased intracranial pressure (including a brain tumor, abscess, hydrocephalus, cerebral edema, and intracranial hemorrhages).

The brain can accommodate a small amount of increased **intracranial pressure (ICP)** initially by expelling CSF and blood into the spinal subarachnoid space. If ICP continues to increase then there is an increased risk of brain damage and herniation.

**Lumbar punctures are contraindicated in the presence of increased ICP because the sudden release of pressure can lead to a cerebral/cerebellar herniation (where part of the brain herniates out of the foramen magnum).*

Signs and Symptoms of increased ICP

Patients with increased intracranial pressure present with **morning headaches** (*since they have been lying flat all night long, the pressure has been building inside their cranium – once they sit or stand upright gravity helps to drain some of the CSF and blood, thus releasing some pressure and improving the headaches*), **vomiting in the morning**, lethargy, irritability, 6th nerve palsy, strabismus, diplopia, and **papilledema**

Infants will also have **bulging fontanel** (*the fontanelles serve as a pop-off valve for the pressure inside of the skull*), suture diastasis, distended scalp veins, and **sun-setting of eyes***, **rapid growth of head circumference**

Definition: **Sun-setting sign refers to the downward deviation of an infant's or young child's eyes as a result of pressure on cranial nerves III, IV, and VI*

CRITICALLY increased ICP

- **CUSHING TRIAD** are very important late findings of critically increased ICP: **increased blood pressure, decreased pulse, irregular respirations**
- Dilation and poor reactivity of one or both pupils is another finding in critically increased ICP

REMEMBER CUSHING'S TRIAD!

↑ BP

↓ HR

Irregular respirations

Diagnosis of increased ICP

The diagnosis of increased ICP is made by monitoring young infants' head circumference and fontanelles, as well as monitoring all patients' signs and symptoms (ie, perform a funduscopic exam to check for papilledema in patients that have signs or symptoms consistent with increased ICP).

What conditions with increased ICP would show a **focal lesion** on a CT scan?

(1) Brain abscess, mass lesion

a. Blockage of CSF, venous outflow or production of CSF

(2) **Hydrocephalus**: a slow evolving syndrome over weeks/months involving pressure from enlarging ventricles and interstitial edema in the periventricular white matter. Its etiology stems from either an obstruction of the CSF along its course of flow or overproduction by choroid plexus papillomas. The symptoms of hydrocephalus can be explained by the following pathophysiology:

a. Stretching of corticopontocerebellar & corticospinal pathways → **ataxia & spasticity**

b. Distention of 3rd ventricle → compression of hypothalamic regions → **endocrine dysfunction**

c. Distention of 3rd ventricle → compression of optic nerves, chiasm and tracts → **visual dysfunction**

Conditions with **Negative CT Scan** and **Abnormal Mental Status**

(1) Hypoxic-ischemic injury, trauma, infection, metabolic derangement or toxic ingestion

(2) **Meningitis** – the pathophysiology of meningitis leading to abnormal mental status and increased ICP includes:

a. Blockage of CSF pathways (leading to hydrocephalus)

b. Toxic cerebral edema from hypoxic ischemic injury

c. Increased cerebral blood flow

d. Multifocal cerebral infarction

e. Most can safely undergo lumbar puncture

i. Contraindications to LPs in patients with presumed meningitis include focal neurologic signs, poorly reactive pupils, and tense fontanel

(3) **Pseudotumor Cerebri** is a **benign** condition leading to increased intracranial pressure that occurs most commonly in **females** of

childbearing age (**typically overweight and obese adolescents**). These patients are NOT critically ill – they tend to be fairly well appearing, although they do often complain of (morning) **headache, diplopia**, & abducens palsy. **Papilledema** is evident on physical examination.

The causes of pseudotumor cerebri are theorized to be diffuse brain swelling vs CSF outflow resistance vs intracranial hypertension. It is most often idiopathic, but can be certain drugs, endocrine disturbances, intracranial venous sinus thrombosis

- a. **Brain imaging is normal**
- b. **Treatment:** acetazolamide, repeated LPs, corticosteroids, surgery

Treatment of increased ICP

There are multiple options in treating increased cranial pressure:

- Elevate head of bed (**reverse trendelenburg**)
- **Mannitol** for osmotic shift
- **Corticosteroids** to reduce edema
- **Acetazolamide & furosemide**: to transiently decrease production of CSF
- **Hyperventilation** induces cerebral vasoconstriction & decreased cerebral blood volume
- **Ventricular catheter** removes CSF & track ICP
- **Pentobarbital induced coma** suppresses cerebral metabolism and blood flow

Psychomotor Retardation

Non-Progressive Psychomotor Retardation

This indicates that the intellectual and physical disabilities do not worsen over time – they essentially remain unchanged as the child ages. Patients with non-progressive psychomotor retardation tend to have very limited thought & physical movements.

One example of this is **static encephalopathy**. Static encephalopathy is typically due to an antenatal/perinatal disturbance or underlying genetic disease. Physical exam reveals no focal neurologic deficits, and lab work is all normal. **Treatment** of static encephalopathy includes infant stimulation & education programs (with the goal of helping the child reach their maximal potential).

Progressive Psychomotor Retardation

If a child's cognitive and physical abilities worsen over time, this poses a much more serious diagnostic dilemma. Children with progressive psychomotor retardation need a **complete laboratory & imaging evaluation**. Some of the potential causes that need to be investigated include:

- Degenerative hereditary & metabolic diseases
- Severe epilepsy
- Chronic drug overdose (lead poisoning, vitamin deficiencies)
- Hypothyroidism
- Indolent brain infections (measles, syphilis)
- Depression

Congenital Malformations of the Brain

These can be due to a variety of injuries during a vulnerable period of gestation (ie, chromosomal, metabolic, genetic, infections, exposure to radiation, drugs, maternal illnesses). Some examples of congenital brain malformations include:

- Congenital hydrocephalus - buildup of excess CSF within the brain
- Holoprosencephaly - varying degrees of failure of the primary cerebral vesicle to divide & expand laterally
- Hydranencephaly - the brain's cerebral hemispheres are absent to varying degrees and the remaining cranial cavity is filled with cerebrospinal fluid
- Schizencephaly - abnormal slits, or clefts, in the cerebral hemispheres of the brain
- Lissencephaly - "smooth brain" - absence of normal convolutions in the cerebral cortex and an abnormally small head (microcephaly)
- Macrogyria & polymicrogyria – large or very small gyri of the brain (can occur in the same brain, or separately)
- Agenesis of the corpus collosum
- Megalencephaly - abnormally large, heavy, and usually malfunctioning brain
- Microcephaly – small head

Disorders Involving Weakness

Weakness

Weakness can originate from 4 places: **upper motor neurons**, **lower motor neurons**, the **neuromuscular junction**, or the **muscles**.

Upper Motor Neuron causes of weakness:

Disorders of the upper motor neurons that lead to weakness include diagnoses like Stroke.

Lower Motor Neuron causes of weakness:

Werdnig-Hoffman, aka Spinal Muscular Atrophy

Spinal Muscular Atrophy is one of the most frequent autosomal recessive diseases (carrier = 1 in 50) involving the progressive degeneration of the anterior horn cells. It may begin intrauterine or any time after and may progress rapidly or slowly. The earlier and more rapidly it progresses, the worse the prognosis. Patients present with **progressive proximal muscle weakness**, flaccid quadriplegia, respiratory failure, and death. The patients' eyes remain bright and engaging throughout the progression of their illness. This is in contrast to patients with myotonic dystrophy.

Idiopathic peripheral neuropathy:

Guillain-Barre Syndrome

In Guillain-Barre Syndrome, the immune system attacks the peripheral nervous system and starts to destroy the myelin sheath that surrounds the nerves. This syndrome often occurs after a respiratory or GI infection. Patients often present with **areflexia** (with normal muscle tone initially), **flaccidity**, and **symmetric ascending weakness**. This condition can resolve spontaneously, but these patients need to be monitored closely while they are symptomatic due to the risk of respiratory failure.

Neuromuscular Junction causes of weakness:

Myasthenia Gravis

The onset of myasthenia gravis is typically during the **teenage years**. It begins with an **acute onset of ptosis**, diplopia, ophthalmoplegia, weakness of extremities, neck, and face. Some children only have eye

findings, while others have such severe symptoms that they require intubation. The symptoms of patients with myasthenia gravis tend to worsen as the day progresses. EMG studies show decremental responses to repetitive nerve stimulation. Myasthenia gravis is caused by antibodies to the acetylcholine receptors at the neuromuscular junction. The treatment for this disorder includes acetylcholinesterase inhibitors (pyridostigmine – is this the same thing as the antibody), removing the thymus gland (this can possibly help through a rebalance of the immune system), immunosuppressive drugs, and supportive care (including intubation if the respiratory muscles are affected).

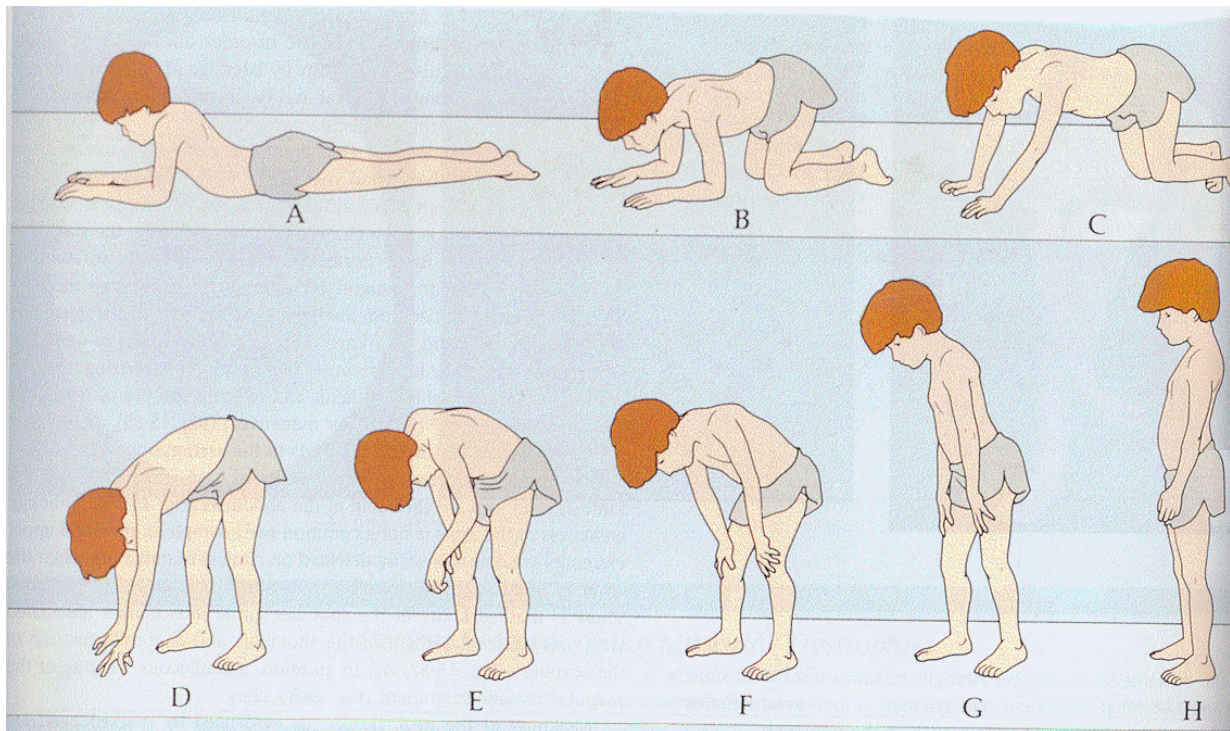
Muscular causes of weakness:

Duchenne's Musculodystrophy

Duchenne's Musculodystrophy is a common **sex-linked recessive** trait with an incidence of 20-30 per 100,000 (~1/3 are due to new mutations). The disease is due to the absence of dystrophin. Dystrophin is a protein associated with the muscle fiber plasma membrane.

Most patient's **develop symptoms around 3 years of age**. The typical child with Duchenne's Musculodystrophy presents with new onset **difficulty running**. The child can also present with **calf pseudo-hypertrophy** (the calf muscles look enlarged, but actually the enlargement is due to fat fibrosis of the gastrocnemius muscles). Other symptoms include **proximal leg weakness**, hyperlordotic and waddling gait, **Gower Sign** (see image below), and arm weakness around 6 years of age.

Most children with Duchenne's Musculodystrophy become wheelchair bound around 12 years of age, and develop serious respiratory issues around 16 years of age. For this disease, laboratory findings reveal markedly elevated serum **creatinine phosphokinase** levels.



Gower's Sign indicates **weakness of the proximal muscles**. The sign describes a patient that has to use their hands and arms to "walk" up their own body from a squatting position due to lack of hip and thigh muscle strength.

Becker Muscular Dystrophy

In Becker Muscular Dystrophy, there is an abnormality in the same gene locus leading to abnormal dystrophin (but NOT absent dystrophin, as is the case with Duchenne's). There is a later onset and a slower progression of the symptoms of the disease.

Myotonic (Muscular) Dystrophy

Myotonic dystrophy is an autosomal dominant disorder involving the gene designated Myotonin Protein Kinase. There are 2 types of myotonic dystrophy.

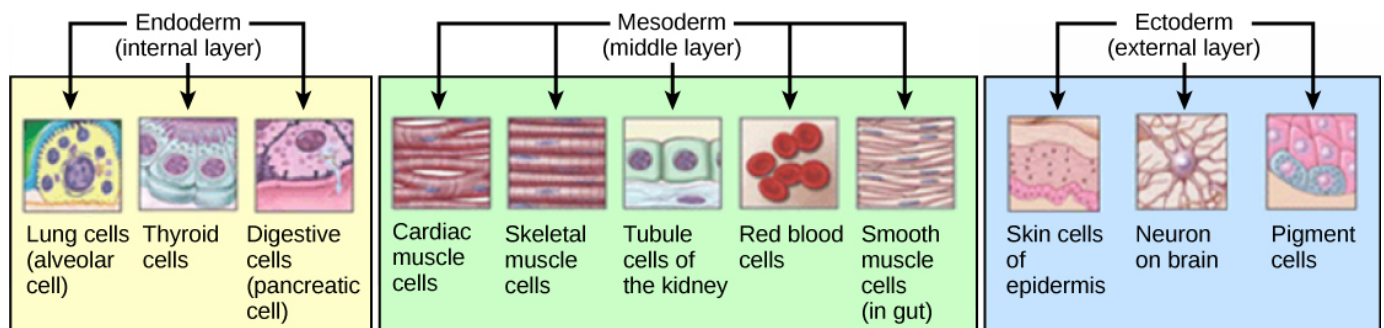
Type 1 – Infant form: These patients will present with **severe generalized weakness** at birth. They will typically also have **expressionless faces**.

Type 2 – Adult form: During adolescence these patients present with **slowly progressive facial and distal extremity weakness**.

Both types present with a **characteristic facial appearance of hollowing muscles, ptosis, and lower lip drooping**.

Neurocutaneous Disorders

The **ectoderm** serves as the precursor to **skin, teeth, hair, nails, and the brain**. Abnormalities of any of these surfaces may indicate abnormal brain development



Tuberous Sclerosis

Tuberous sclerosis is an autosomal dominant disease with an incidence of about 1 in 10,000.

Patients present with hamartomas on various organs such as the brain, eye, skin, kidneys, and heart. Patients also present with **facial angiofibromas, intellectual disability & severe epilepsy.**

The cutaneous findings that are strongly associated with Tuberous Sclerosis include:

- **Shagreen patches** (orange peel)
- Ungual fibromas
- Retinal hamartomas
- **Ash-leaf spots** (hypopigmented macule)

A shagreen patch is a firm yellowish-red or pink area of nodules slightly elevated above the surrounding skin and often described as having the texture of an orange peel.

The ash-leaf spot is the earliest skin lesion in patients with tuberous sclerosis. This hypomelanotic macule is most easily visualized with a Wood's light. Ash-leaf spots are found at birth or in early infancy in 87 percent of patients with the disorder. By themselves, these hypomelanotic macules are not diagnostic of tuberous sclerosis because they are also relatively common in the general population

Ungual fibromas are flesh-colored papules that may displace and distort the nail bed.

If a physician notices these cutaneous findings they may want to investigate further to see if the patient has Tuberous Sclerosis and is at risk for further lesions on their brain (and associated symptoms).

Cutaneous Findings in Tuberous Sclerosis

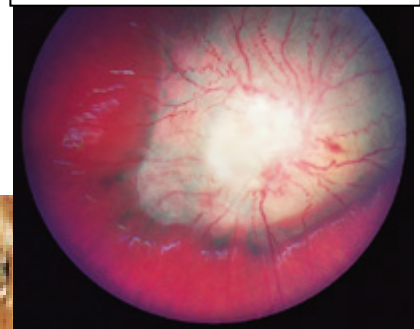
Shagreen Patch



Ash Leaf Spot



Retinal Hamartoma



Facial Angiofibromas



Ungal Fibroma



Neurofibromatosis 1 (NF1)

Neurofibromatosis is an autosomal dominant disease with an incidence of about 1 per 3,000. It is caused by a mutation of the NF1 gene on chromosome 17q11.2 which codes for the tumor suppressor gene neurofibromin (a major negative regulator). In about 30-50% of cases there is a new mutation that causes the disease.

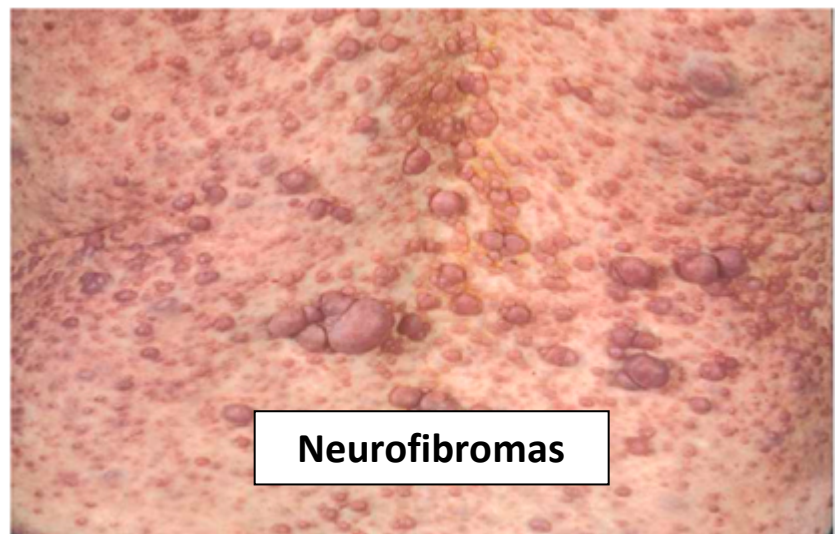
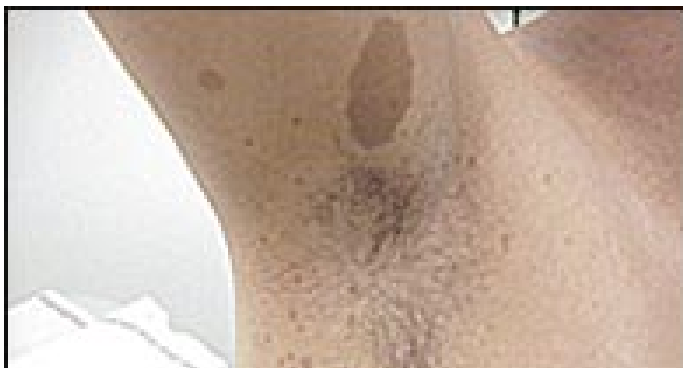
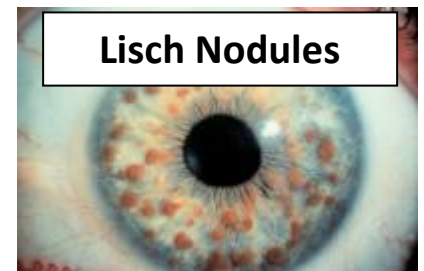
The **tumors** in patients with NF1 **grow in the nervous system**. The most common tumors are **neurofibromas** or tumors of the peripheral nerves.

Patients present with **learning disabilities, scoliosis, macrocephaly, headache, and optic gliomas**. **The most common cause of death is malignancy**. The average life expectancy for individuals with this disease is decreased by 10-15 years.

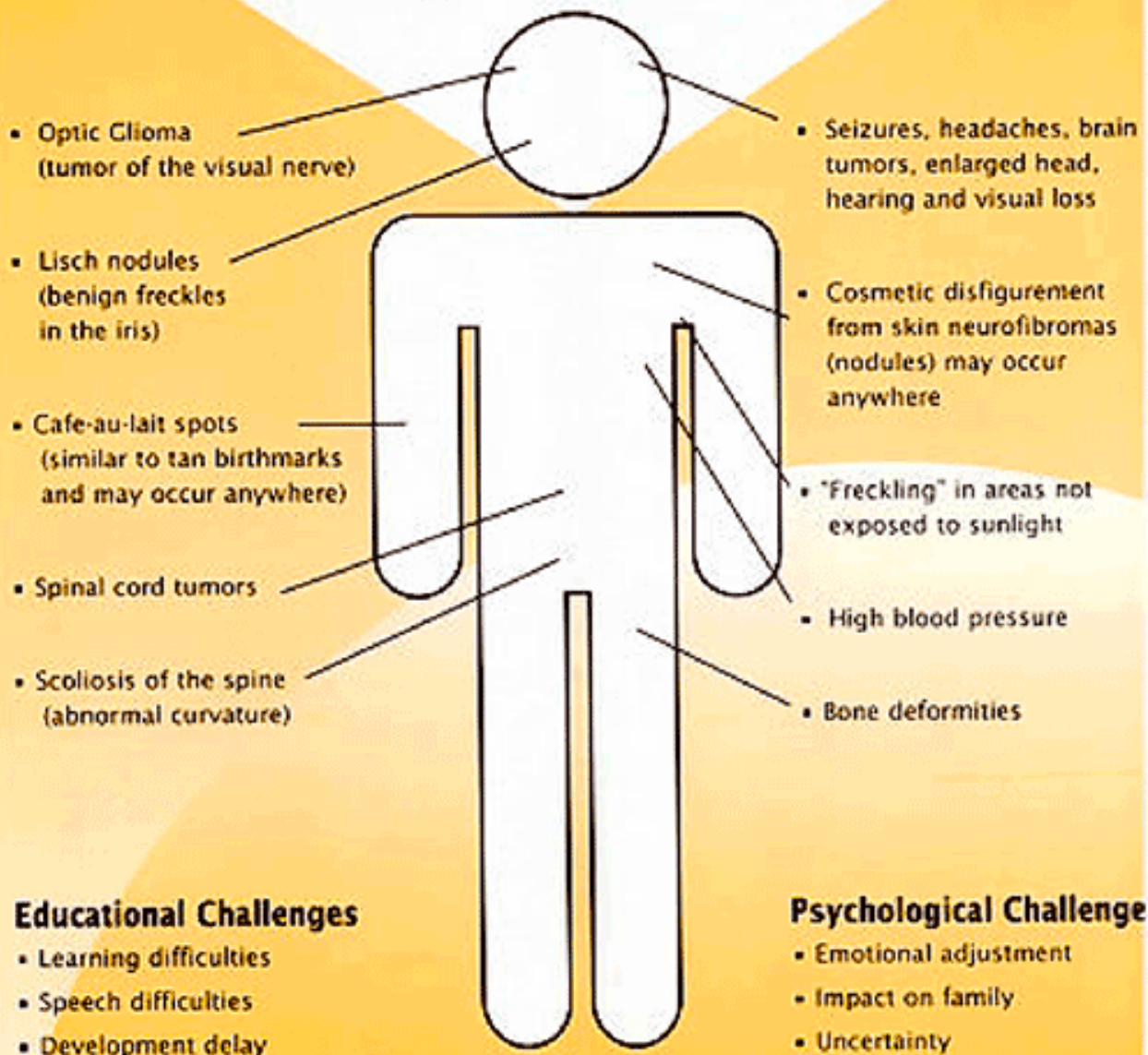
Diagnosis of NF1

To diagnose neurofibromatosis, patients must present with at least two or more of the following:

- **≥ 6 “café-au-lait” spots** (well demarcated, flat hyperpigmented lesions (“birthmarks”) with sharp borders)
 - >5 millimeters in diameter in children, >15 millimeters in adolescents and adults
- **≥ 2 neurofibromas** (small, rubbery growths)
- **Freckling in the area of the armpit** or the groin
- **≥ 2 Lisch nodules** or iris hamartomas
- An optic nerve glioma
- **Scoliosis**, abnormal development of the sphenoid bone, or the tibia
- **A parent, sibling, or child with NF1**



Possible Signs of Neurofibromatosis



While it is unlikely that any one person diagnosed with NF will experience all of these symptoms, it is difficult to predict the severity or progression of the disorder in any individual case.

Neurofibromatosis Type 2 (aka, Schwannomatosis)

Neurofibromatosis (NF2) or Schwannomatosis is an autosomal dominant disease with an incidence of about 1 in 33,000. About half of these cases do not have a family history of the disease.

Schwannomatosis is due to a mutation in the tumor suppressor NF2 gene on chromosome 22.

This disease predisposes individuals to **multiple intracranial and spinal tumors**. By the age of 30 years, almost all affected individuals develop **bilateral vestibular schwannomas**. Individuals may also develop schwannomas of other cranial and peripheral nerves, meningiomas, ependymomas, and astrocytomas. Cataracts are also common. With this disease there are **NO lisch nodules, café-au-lait spots, or axillary freckling**.

The average lifespan for patients is about 40 years.

Diagnosis of NF2

The diagnosis for Schwannomatosis can be done in multiple ways:

- (1) **B/L vestibular schwannomas**
- (2) **Family history of NF2** (parent, sibling, or child) plus a unilateral vestibular schwannoma before age 30
- or**
- (3) **Any two of the following:**
 - o glioma
 - o meningioma
 - o schwannoma
 - o juvenile posterior subcapsular/lenticular opacity (cataract) or juvenile cortical cataract

Sturge-Weber Syndrome

Sturge-Weber syndrome is a sporadic syndrome rather than a genetic one.

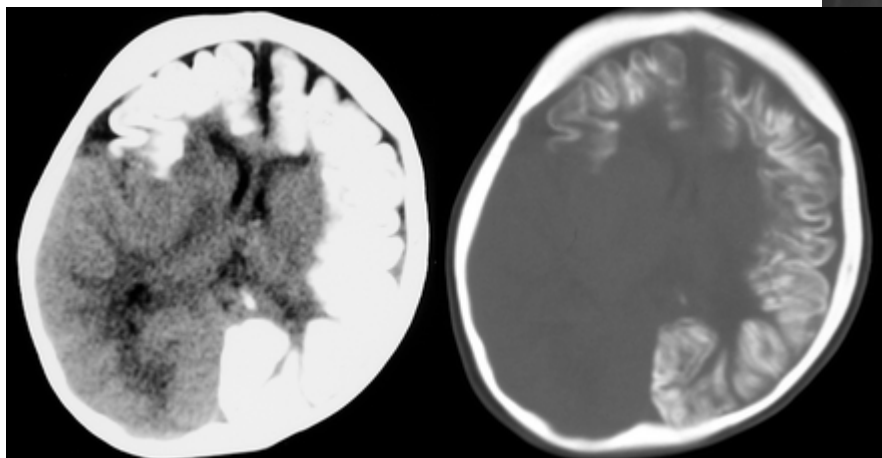
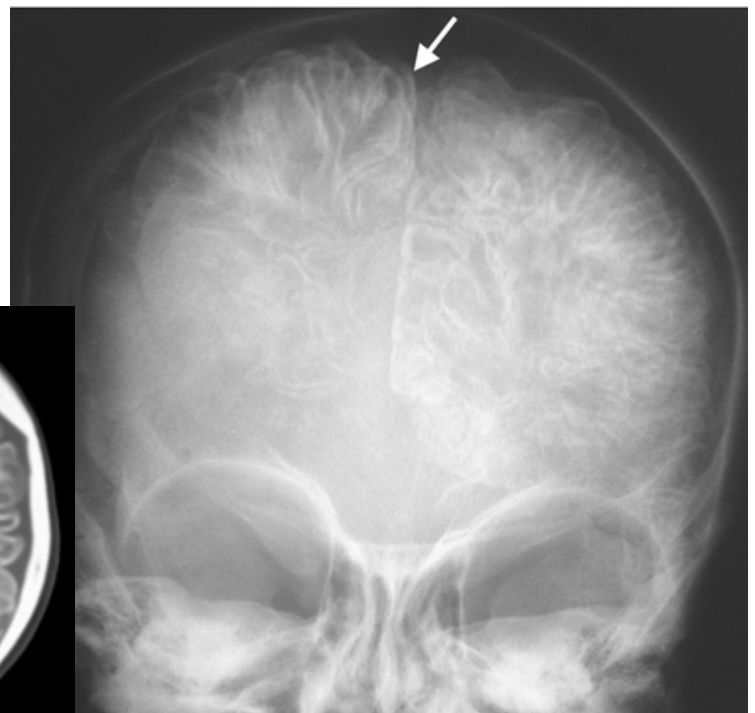
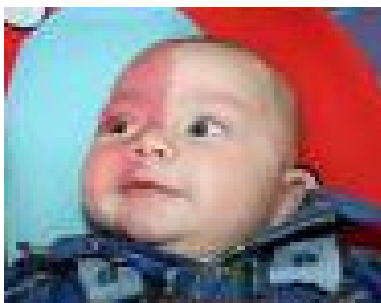
Patients present with the following symptoms:

- **Angiomas of the leptomeninges**
- **Ipsilateral facial port-wine nevus** (*see images*)
 - ***Covering part of the forehead and upper eyelid***
- 30-50% of patients have glaucoma
- **Seizures**
 - ***Due to ischemic injury to the brain underlying the meningeal angiomas***
- Progressive ischemia can develop
 - Leading to hemiparesis, intractable epilepsy, & dementia



Calcium is detected in the gyri of the brain underlying angiomas. This gives a “tram track” appearance on imaging (*see pictures below*).

Treatment includes anticonvulsants, hemispherectomy, and laser surgery for cosmetically improving the look of the facial nevus.



References

- Kliegman RM. *Nelson: Essentials of Pediatrics*. 7th Edition. Saunders; 2014: 1328-1458
- Osborn LM, DeWitt TG, Lewis RF, et al. *Pediatrics*. Elsevier Mosby; 2005: 1162-1196
- National Institute of Health. National Center for Biotechnology Information Website. <http://www.ncbi.nlm.nih.gov/bookshelf/> . Accessed November 20, 2017.
- National Institute of Health. National Institute of Neurologic Disorders and Stroke Website. <http://www.ninds.nih.gov/disorders>. Accessed November 20, 2017.

Test Yourself on This Material

Quizlet: <http://quizlet.com/10ejsp> =
**Pediatric Neuromusculoskeletal
Disorders Part 1**